DUNCAN MCGOUGH

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EXPERIENCE

Space Exploration Technologies Corporation (SpaceX) June 2020 - Present
Software Engineer (2024), Propulsion Engineer (2020) *Hawthorne, CA*

sx_fluids Python Library
- Created widely-used thermal fluids simulation Python library, `sx_fluids`, to model real/multiphase fluids
- Developed CI/CD/K8s pipeline and monorepo with automated testing and coverage analysis, including extensive documentation
- Trained analysts and team members on software engineering best practices, allowing for regular contribution

Rust Thermofluids ODE Network Solver, "Reef"
- Built and expanding upon next-generation thermal-fluid nodal network transient solver for analysis org, complimenting `sx_fluids`

Starship Tank Press Model
- Developed primary tank pressure simulator, now essential for flying Starship
- Leverages `sx_fluids`, and has features for easy analyst accessibility and high-performance computing (HPC) tooling

Rust Thermophysical Property Library Tools
- Developed thread-safe wrappers for REFPROP (includes Python API), improving performance by several orders of magnitude
- Created high-performance Peng-Robinson equation-of-state library with robust error handling to compliment REFPROP

Thermal Fluids Webapp Ecosystem
- Created Kubernetes-hosted webapp suite and ecosystem for performing quick trades and calculations, saving time for analysis teams
- Designed ecosystem for team contribution, which has now been adopted and is regularly-used

CFD Models
- Built multiphase slosh models for Starship ascent and microgravity phases, optimizing mass-to-orbit and mission success
- Created multiphase vapor-pullthrough models to minimize propellant residual by several tonnes and assess flight risk
- Constructed highly-compressible gas simulations for RCS thrusters and tank-internal press gas diffusers

Space Exploration Technologies Corporation (SpaceX) Summers 2018, 2019
Satellite Development Intern (2018), Associate Engineer - Post Grad, Thermal-Fluid Analysis (2019) *Hawthorne, CA*
- Developed Starlink hall-effect thruster thermal model to determine duty cycle and improve performance. Reduced bus mass.
- Analyzed Dragon spacecraft Draco hypergolic engine, bounding fault cases for propellant unsettling
- Created satellite thermal models of star trackers, propellant tanks, and propulsion avionics, used by qualification team
- Analyzed Falcon 9 S2 fuel baffle leakage; discovered unacceptable leakage

Roccor LLC May 2017 - August 2017
Thermal Engineering Intern, Thermal Group *Longmont, CO*
- Engineered thermal vacuum chamber for deployable composite CubeSat radiators. Met NASA/Roccor project requirements, budgets, and deadlines. Advised manufacturing of chamber. Designed and simulated, validating with test data.

COBRA "Spaceshot" Suborbital Rocket Development Team March 2016 - January 2018
Propulsion Lead *Boulder, CO*
- Manage propulsion subteam. Modeled oxidizer injector assembly and used CFD and FEA to validate designs. Calculated nozzle geometry and flow rates. Designed liquid oxidizer storage and delivery/injection systems for a hybrid rocket.

Gridded Electrostatic Ion Thruster Research BalloonSat Project January 2016 - May 2016
Team Lead *Boulder, CO*
- Led a team to design, test, and fly a high-altitude BalloonSat payload. Developed, researched, and constructed a low-cost gridded electrostatic ion thruster to test at altitude for deployment upon CubeSats.

EDUCATION

University of Colorado, Boulder August 2015 - May 2020
Engineering Honors Program, Chancellor's Scholarship
Masters of Science, Aerospace Engineering Sciences with focus on Fluids and Propulsion — May 2020
Bachelor of Science, Aerospace Engineering Sciences — May 2019
Bachelor of Music, Violin Performance (College of Music Scholarship Award) — August 2015 - May 2016

STRENGTHS AND SKILLS

Languages	Rust, Python, MATLAB, Julia, C++
Frameworks	Docker, Kubernetes, Maturin/PyO3, Redis, Leptos, Clap (Rust CLI), Tokio, Axum, Serde, Flask, Traefik
Software	STAR-CCM+, OpenFOAM, ANSYS, Thermal Desktop, CAD (SolidWorks, NX)
OS	Linux, OSX, Windows, slurm (HPC), ProxMox, using Virtual Machines
Personal Interests	Violin, freeride skiing, mountain biking, martial arts, spikeball, onewheeling, FPV drones